Allocating What Programs Touch: Non-Binding Per-Site Reservations from Measured Memory Footprints

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Abstract

A program asks the allocator for the memory it might need, and the operating system and a cluster scheduler must honour every request, although the program often touches only part of it. We use MemPrint, a Pin-based footprint profiler, to measure what each allocation site requests and what the program touches in each block, and we pass these measurements to the compiler and the runtime as *non-binding reservations*. The system has three parts. First, a per-site mode of the Pin tool records each large block's touched bytes, highest touched offset and touched page ranges, the calling context that allocated it, and the growth of every container; zeros written over untouched memory do not count as use. Second, `MemPrintResize`, an LLVM 21 pass plugin, routes every allocation through a runtime with a site identifier that survives edits elsewhere in the program, raises the capacity of growing `std::vector`s, strings and hash tables per call site, with element sizes from the debug information, and can embed the reservations in the binary. Third, `libmpr`, linked or preloaded, commits only the predicted prefix or ranges of a block and maps the rest `MAP_NORESERVE`, so that an under-prediction costs commit accounting and never correctness; it also skips the zeroing of pages a program has never touched. Every program we ran produced its original output. On llama.cpp, which sizes and zeroes a KV cache for an 8192-token context, `libmpr` reduces peak commit by 86% and peak RSS by 54% with a 135M-parameter model (90% and 45% with a 360M model), and profiles from other model sizes do as well. It reduces commit by 85% on llama2.c, whose KV caches are used in one range per layer, by 47% on darknet AlexNet inference, and by 22% on miniVite with the pass's per-call-site container capacities. Across a 23-program HPC and interpreter corpus, where most programs already hold about what they touch, it changes no output and no total RSS, keeps run time within −5% to +7%, and reduces total peak commit by 10.7%. The pass adds 0 to 3% to compile time, and 96.6% of its site identifiers survive an edit that changed 19% of the previous ones.

1 Introduction

A program's memory request and its memory use are different quantities, and every layer below the program plans for the request. The allocator maps what is requested, the kernel charges the mapping to the commit limit [21], and a batch scheduler on a cluster with disaggregated memory reserves the request, or an estimate of it, before it places the job [45, 19]. Our earlier work built MemPrint, a Pin-based profiler [26, 31] that measures the bytes a program touches over its run, and showed that a scheduler which knows a job's footprint curve rather than its request can pack jobs much more tightly [30]. However, a scheduler that reserves less than a program requests does

not change what the program asks for. The program still allocates the full request, and the kernel still charges it.

The gap between request and use is large in some programs. darknet [37] running AlexNet inference commits 534.7 MB and touches 270 MB, because it allocates training-only weight-update buffers that inference never uses. llama.cpp [11] sizes its KV cache for the model's full 8192-token context and then writes zeros over all of it, so a 32-token completion holds 189 MB of cache in RAM and uses 0.4% of it. miniVite's edge list grows by doubling, so for a moment the old and new buffers are both live [12].

Therefore, we ask whether a measured footprint can be passed down to the allocator, so that the program obtains what it touches rather than what it requests, without trusting the measurement for correctness. Specifically, we treat MemPrint's per-site measurements as *non-binding reservations*. A block always receives every byte the program requested, but only the predicted part is committed, and the rest is mapped so that the kernel backs it only if the program touches it. A wrong prediction then costs accounting and never causes a fault the program can observe. *To the best of our knowledge, this is the first compiler pass and runtime that commit only the profiled part of each heap block, leaving the rest to be backed on demand.* Existing profile-guided allocation work, such as LLVM's MemProf, changes where an object is placed but commits every requested byte [25], and existing container-sizing tools give advice to the programmer or replace data structures in managed runtimes [10, 28] (Section 3).

In this work, we make the following contributions:

- We extend MemPrint's Pin tool to profile requested against touched bytes per allocation site, with each block's touched bytes, highest touched offset and touched pages measured exactly when the block is freed (Section 5).

- We extend the profile with calling contexts, so that a site reached through an allocation wrapper is split by its callers, with touched page ranges for blocks used sparsely across their length, with container growth chains, and with zeroing of untouched memory left out of use (Section 5, Section 6).

- We present `MemPrintResize`, an LLVM pass plugin that gives every allocation call a site identifier that survives edits elsewhere in the program, routes address-taken allocators through thunks and large zeroing through the runtime, and raises the capacity of growing vectors, strings and hash tables per call site, with element sizes from the debug information (Section 7).

- We present `libmpr`, a runtime that commits a block's predicted prefix or ranges and maps the rest `MAP_NORESERVE`, matches reservations by calling context, grows blocks by moving page tables, skips the zeroing of pages that were never touched, and refuses address-keyed reservations profiled on another build (Section 8).

- We evaluate the system on darknet, miniVite, two LLM inference engines and a 23-program corpus. It cuts llama.cpp's commit by 86% and its resident memory by 54%, and llama2.c's commit by 85%, with identical outputs and no time cost, and we measure what each refinement adds (Section 9).

- We report what the measurements show about footprint estimation: where miniVite's extra memory comes from, why the bytes a program touches can exceed its resident set, and why the time axis of a footprint curve matters more than its unit (Section 10).

2 Background

2.1 Three measures of a program's memory

We use three measures throughout, and the results depend on which one is meant.

Bytes touched. MemPrint's footprint is the number of distinct bytes the program has read or written in memory that is still live, measured from every memory reference under Pin [31]. With `-track_frees`, a freed range leaves the footprint. Bytes touched are a property of the program and its input, and they do not depend on the allocator or on the page size.

Resident set (RSS). The kernel counts the pages of a process that are in RAM. RSS includes file-backed pages, such as a memory-mapped model file and the code of every shared library, and it counts a transparent huge page as 2 MB [22]. We sample it from `/proc/<pid>/smaps_rollup` every 20 ms and report the peak sample. RSS is the quantity the kernel reclaims and the out-of-memory killer acts on, and it is the "RSS truth" of the scheduling study [30].

Commit. Linux charges private writable mappings to the commit limit when they are created [21]. We measure commit as the total size of the mappings that `/proc/<pid>/smaps` flags `ac` (accountable), sampled with RSS. A mapping created with `MAP_NORESERVE` is not charged. Under the default heuristic overcommit policy commit is only advisory, but under strict overcommit (`vm.overcommit_memory=2`) it is a hard limit, and in both cases it is the memory the system has promised to the process.

2.2 Why untouched memory is often free already

Linux backs anonymous memory lazily. A large `malloc` or `calloc` block is served by `mmap`, and a page of it becomes resident only when the program first writes it, or reads it, in which case the kernel maps a shared zero page that RSS does not count. Therefore, a block that the program never touches costs commit but no RSS. darknet's 250 MB of unused weight-update buffers are an example. Reducing RSS requires a change in how memory is obtained, not a smaller reservation. In our measurements RSS fell only where the program, its allocator, or the kernel touched pages the program did not need: vector growth that keeps two buffers live, a zeroing `memset` over fresh memory, and transparent huge pages that back 2 MB on the first touch.

3 Related work

We searched the compiler, allocator, operating-system and LLM-serving literature for an existing pass that resizes heap allocations from a profile of what a program touches. To the best of our knowledge, no prior compiler pass commits only the profiled part of a heap block and leaves the rest to be backed on demand. The closest work falls into five groups, which Table 1 summarises.

Profile-guided compilation of allocation sites. LLVM's MemProf is the closest existing compiler work [25]. It records, per allocation context (a call stack of function, line and column), the allocation count, access counts at 64-byte granularity and lifetimes. Its `MemProfUse` pass matches the profile to allocation calls through debug information, clones functions under LTO to separate contexts, and rewrites `operator new` into a variant that carries a hot or cold hint, which an allocator such as TCMalloc uses to place the object. `MemPrintResize` shares this structure: an instrumented profile, sites matched through the inlined debug location chain, and allocation calls rewritten to a runtime that accepts site information. However, MemProf changes only where an object is placed and leaves every requested byte committed, whereas `MemPrintResize` and `libmpr` change how much of each block is committed and how it is backed, and a site's reservation is a size

System	Profile	Action	Automatic	Wrong profile
LLVM MemProf [25]	access density and lifetime per allocation context	hot or cold hint to `operator new` (placement)	yes (compiler + TCMalloc)	placement only
libstdc++ profile mode, Perflint [23, 10]	container operations per call stack	advice: initial size, container choice	no (programmer)	–
Chameleon, Brainy [40, 14]	collection usage	select a collection implementation	partly	slower or larger collection
Makor et al. [28]	data-structure use per allocation site	replace with a smaller implementation (GraalVM Native Image)	yes	guarded by the implementation
Odaira (IBM patent) [32]	highest index accessed per allocation context	JIT sets the initial array size	yes (Java JIT)	array grows by copying
Xu et al., DHAT [42, 43, 41]	cost and benefit, or bytes accessed, per site	report bloat	no (programmer)	–
This work	bytes touched, highest offset and touched page ranges per block, site and calling context; container growth chains per call site	commit the predicted prefix or ranges, the rest `MAP_NORESERVE`; raise container capacities per call site; skip zeroing untouched pages	yes (pass or preload)	costs commit accounting only

Table 1: Profile-guided work on heap allocations, and how a wrong profile affects each.

model rather than a temperature. MemProf's profile is collected by a sanitizer-style runtime in the program itself, while ours comes from MemPrint's Pin tool, which needs no recompilation for the profiling run and measures bytes rather than 64-byte granules.

Profile-guided container sizing and selection. Perflint traces C++ standard library operations and reports where a better use exists [23], and libstdc++'s profile mode, its descendant, prints advice such as a vector's suggested initial size or a vector that is never filled to its reserved size [10]. The advice is for the programmer to apply, and libstdc++ notes that it depends on the profiled input. Chameleon selects Java collection implementations from runtime metrics [40], Brainy selects C++ data structures with machine-learned cost models [14], and Makor et al. automatically replace data structures at allocation sites in GraalVM Native Image with versions specialised to their profiled use, which reduces memory by up to 13.85% (2.94% on average for microservices) [28]. The closest to our vector capacity is an IBM patent by Odaira, in which a Java JIT records the highest index accessed per allocation context and sets the initial array size from it when it inlines the allocation site [32]. Our capacity reservation differs in that it can only raise the capacity a growing vector would have chosen, so a wrong prediction cannot shrink a block below its request, and it applies to C++ compiled ahead of time, with the profile taken from an unmodified binary.

Bloat detection and heap profiling. Xu et al. find Java data structures whose cost exceeds their benefit [42], and Xu and Rountev find containers that are overpopulated or underused [43]. Heap profilers that report lag, drag and void, the time a block is allocated but not yet or no longer used [38], and DHAT, which reports per allocation stack how many bytes were read and written and flags blocks that are not completely filled in [41], also diagnose over-allocation. All of them leave the fix to the programmer. MemPrint's `-sites` profile reports similar quantities, but the system acts on them without a source change.

Placement, compaction and huge pages. Mesh compacts the heap of unmodified C and C++ programs by merging pages whose live objects do not overlap [35], Maas et al. predict object lifetimes from allocation call stacks to place long-lived objects together [27], and HALO groups heap objects by profiled access affinity after link time [39]. These systems use per-site profiles to change placement, whereas we use them to change commitment and backing. jemalloc and TCMalloc's hugepage-aware Temeraire return freed memory to the kernel more eagerly than glibc [7, 13], but they still commit every requested byte. Ingens and HawkEye showed that transparent huge pages inflate the resident set of sparsely used memory, and they promote a region to a huge page only when enough of it is used or deduplicate zero-filled huge pages in the kernel [17, 34]. `libmpr`'s huge-page policy addresses the same bloat from user space, for the specific pages whose zeroing it skipped. Yang et al. measured the cost of zeroing new memory in managed runtimes and made zeroing cheaper with non-temporal and concurrent zeroing [44]. `libmpr` instead avoids writing zeros to pages that are already zero, which saves resident memory as well as time.

Calling contexts, stale profiles and partial commitment. Several ideas we build on come from profiling and operating systems. Calling-context profiling attributes a measurement to the chain of calls that led to it, recorded as a calling context tree [2] or encoded cheaply as a probabilistic calling context value per thread [4], and MemProf disambiguates allocation contexts by cloning functions under LTO [25]. We split a site by its caller frames, taken with the call frame information at allocation time, which is enough to separate the callers of an allocation wrapper. Sample-based profile-guided optimisation keys its profile by the line offset from a function's first line so that a profile outlives edits elsewhere in a file [5, 24], and stale profile matching recovers a profile for a changed binary [3]. We use the offset scheme for site identifiers, and GNU build identifiers [9] to refuse address-keyed reservations on another build. Windows lets a program reserve address space and commit chosen pages of it later [29], and Linux's `MAP_NORESERVE` leaves a mapping out of the commit charge [21]. Our shaped reservations commit chosen ranges of a block from a profile, without a change to the program.

LLM serving and memory pools. PagedAttention allocates an LLM server's KV cache in fixed-size blocks as tokens arrive [16], which requires paged attention kernels and a memory manager in the serving framework. vAttention keeps the KV cache contiguous in virtual memory and backs it with physical GPU memory on demand through the CUDA virtual memory API, which removes the kernel changes but still requires changes to the serving framework [36]. Our llama.cpp result obtains the same kind of on-demand backing on the CPU for an unmodified engine: the cache keeps its full virtual size, and only the pages it writes are backed, because `libmpr` skips the zeroing that would otherwise touch the whole cache. At cluster scale, Pond predicts how much of a virtual machine's memory will stay untouched and places that part on a CXL pool [19], and Zacarias et al. show that a 60% overestimate of memory demand reduces the throughput of a disaggregated system by about 25% [45]. These works motivate knowing what a job touches, which our earlier study measured for scheduling [30] and this work passes to the allocator.

4 Overview

Figure 1 shows the pipeline. First, the program runs once under MemPrint's Pin tool with `-sites`, which writes one row per large block. Second, `sites.py` fits each site's reservation from one or more such runs. Third, the reservations reach `libmpr`. A program can use the system in two ways. In the *preloaded* configuration, `libmpr` is loaded with `LD_PRELOAD` into an unmodified binary, a site is the return address of the allocation call, and the reservations come from a file. In the *pass* configuration, the program is built with `MemPrintResize`, a site is a hash the compiler computes,

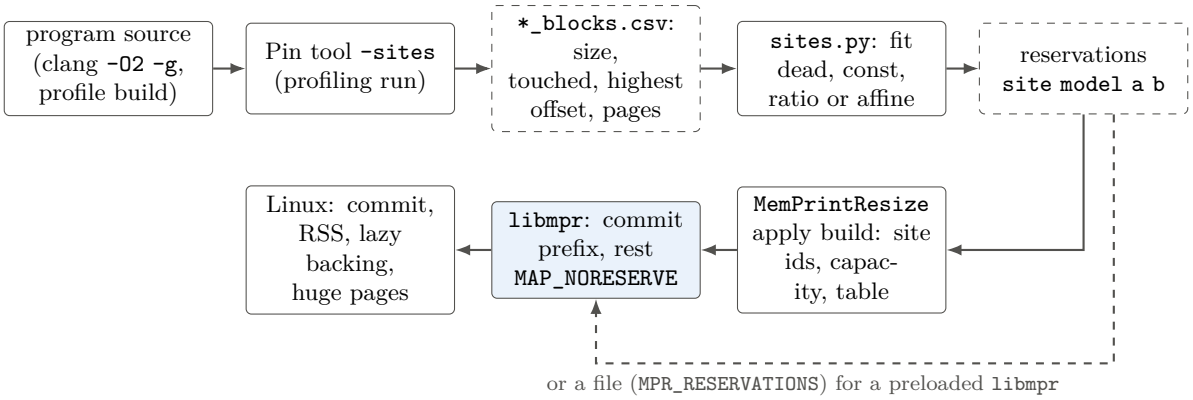


Figure 1: The pipeline. A profiling run of the program under MemPrint's Pin tool measures each large block; `sites.py` fits one reservation per allocation site; the reservations reach `libmpr` either embedded in a binary built with `MemPrintResize` or from a file when `libmpr` is preloaded into an unmodified binary.

and the reservations can be embedded in the binary. Only the pass configuration can change a vector's growth capacity, because the capacity is computed inside inlined library code, before any allocation call.

5 Per-site profiling in MemPrint

The `-sites` mode of the Pin tool runs with the exact splitter, `-track_frees` and the bytes-touched footprint, which keeps one bit per byte for every touched page. It records, for every large block, what the program asked for and what it did with it, and it writes four files: `*_blocks.csv` (one row per block of at least 64 KB), `*_sites.csv` (totals per site), `*_capacity.csv` (container growth, Section 5.4) and `*_images.csv` (the images that site keys name, with their paths).

5.1 Sites and calling contexts

The tool hooks the allocator's entry points (`malloc`, `calloc`, `realloc`, the aligned variants and every `operator new`). For the outermost allocator call of a thread it records the call's return address, written as the image name plus the address in the image file (for example `llama-completion+0x92266f`), which stays the same across runs despite address-space layout randomisation. Because `operator new` is hooked as an allocator, an allocation made by inlined standard-library code resolves to the program's own code. In a program built with `MemPrintResize`, the tool also hooks `__mpr_alloc` and the other runtime entry points, reads the site identifier from the argument the pass added, and names the site `id:0x...`.

A site is a single call, and a call inside an allocation wrapper serves every caller of the wrapper. llama.cpp allocates its KV cache, its compute buffers and its weights' scratch space through one function, `ggml_aligned_malloc`, so all of them share one site, although the program uses them very differently. Calling-context profiling distinguishes the callers of a function by the chain of calls that reached it [2, 4], and LLVM's MemProf records allocation contexts as call stacks for the same reason [25]. With `-site_context` k (2 by default), the tool records, for every allocation of at least 64 KB, the k return addresses above the allocation call. It takes them with `PIN_Backtrace`, which unwinds through the call frame information, so the program needs no frame pointers, and `libmpr` takes the same frames with glibc's `backtrace()` at run time. The frames are written with the

block as image keys, `frame1|frame2`. Backtraces are taken only for large requests, so a program's many small allocations add no unwinding cost.

5.2 Block summaries

When a release (a `free`, a `realloc` that moves, or an `munmap`) is applied, the tool is about to clear the block's bits from the footprint. Just before it does, it reads the block's touched bytes, the offset of its highest touched byte and its number of touched pages from the per-page bitmaps, and it writes the block's touched pages as runs, `first:count` relative to the block's first page, merging the closest runs when there are more than 64. Releases are queued in order with each thread's buffered references, so a summary sees every reference the program made to the block before freeing it. Blocks still live at exit are summarised at exit. `calloc`'s own zeroing of a reused heap chunk is measured by a probe queued at the `calloc` exit and recorded separately as `alloc_touched`.

5.3 Zeroing that does not count as a touch

A program that clears a buffer it has just allocated writes every byte of it, so a bytes-touched profile counts the whole buffer as used. llama.cpp clears its 189 MB KV cache this way and then writes a few rows of it, so the first profile reserved all of it. Writing zeros over memory that the program has not yet touched does not change what the memory holds, and libmpr skips such writes (Section 8.5). With `-skip_zeroing`, on by default with `-sites`, the tool leaves these writes out of the footprint. It hooks the entry of every `memset` and `bzero` implementation in the C library (skipping the IFUNC resolvers that select among them, which the loader calls once), libmpr's `memset` and the pass's `__mpr_zero`. At each entry it queues, in order with the thread's buffered references, the range that the call clears: $[p, p + n)$ when the value is zero and $n \geq 4$ KB, and an empty range otherwise. Stores made by the code of those routines are tagged when they are instrumented, and a tagged store into the current range counts as time but does not set the footprint's bits. A later read or write by the program touches the bytes as usual. A byte that was already touched stays touched, so the option changes only what zeroing of untouched memory contributes. Yang et al. measured that zeroing new memory costs managed runtimes 2.7 to 4.5% of their time on average and made it cheaper [44]. Here the point is that zeroing of untouched memory is not use.

5.4 Container growth chains

In a program built with `MemPrintResize`, a container that grows calls `__mpr_capacity` before it allocates its new buffer (Section 7.3). The tool hooks that call and `__mpr_growth`, and it attaches the capacity key, the call site that grew the container, to the buffer that the next allocation returns. Buffers with the same key form a growth chain when each new buffer is larger than the previous one and is allocated while the previous one is still live, which is how `std::vector`, `std::string` and hash tables grow. For each key, `*_capacity.csv` records the number of chains, the number of chains whose final buffer stayed below 64 KB and the largest of those, and, for the other chains, the final buffer's size and its highest touched offset, which is the bytes the container used.

5.5 Images and builds

An image key is an address in one build of an image and names unrelated code in another build. The tool writes the full path of every image it saw, and `sites.py` records each image's GNU build identifier, a hash that the linker writes into a note section [9], with the reservations (Section 8.6).

Tests. The suite checks exact values on programs written for the purpose: a dead block, a block touched only in a prefix, a block touched only at its tail, a `calloc` on a reused chunk, a `realloc` growth chain, a growing vector, an allocation wrapper called from two functions that use their blocks differently, and a 1 MB buffer cleared with `memset` and then written in its first 64 KB (touched: 64 KB).

Two further options. In the course of this work we also added `-clock seconds`, under which the timeline's snapshots are taken every N milliseconds of wall time and every row carries a `Seconds` column, and `-footprint pages`, which counts the 4 KB pages that hold a touched byte. We use both in Section 10.1.

6 Fitting reservations

`sites.py` reads the blocks of one or more profiling runs, keeps the blocks of at least 64 KB (the ones `libmpr` serves), and fits, per site, the bytes a block will need as a function of its requested size n.

Prefix models. `libmpr` commits a block from its start, so a prefix prediction is safe for a block only if it reaches the block's last touched byte, and the fit uses the highest touched offset. Four models are considered:

- `dead`: 0, when no block of the site was touched;
- `const`: a, when the highest offset does not grow with the request;
- `ratio`: $a\,n$;
- `affine`: $a\,n + b$, the lowest line through two blocks that lies above the covered blocks.

A model *covers* a set of blocks if its prediction reaches each block's highest offset. Among the models that cover blocks holding at least a quantile q of the site's requested bytes ($q = 0.9$ by default), the fit keeps the one that commits the least over the training blocks. A site is left out when its best model would commit at least 95% of what was requested.

Shaped reservations. A block that is used sparsely across its whole length defeats a prefix. llama2.c's KV caches are laid out as [layer][position][dimension], and a run of 128 tokens writes the first rows of every layer, so the highest touched offset lies at 93% of the block while 12.5% of it is used. The `runs` model commits a set of ranges of the block, given as fractions of its size: the union of the training blocks' touched page runs, with the closest ranges merged until at most 32 remain. Windows separates reserving address space from committing it, so that a program can commit only the pages it will use in a reserved region [29], and the `runs` model applies the same split, chosen from a profile. The fit keeps `runs` when it commits less than 70% of the best prefix model and saves at least 64 KB.

Calling contexts. When a site's blocks have caller frames, the fit groups them by their first caller frame, then by their first two, and fits each group separately. It keeps the shallowest grouping whose total commitment is lower than the merged fit's by at least 5% of the site's requested bytes. The site then gets one line per context, `site@frame1[@frame2] model a b`, and its merged line, if it saves anything, serves contexts that the profile did not see.

Container capacities. For each capacity key with chains whose final buffer reached 64 KB, the fit writes `key cap A B`: A is the q-quantile of the final buffers' used bytes, and B is the largest final size of the key's small chains (0 when there are none). `libmpr` reserves A bytes once a container asks for more than B, so that the many small containers that a call site may grow keep their usual growth.

Code	Becomes	Purpose
`malloc`, `calloc`, `realloc`, `aligned_alloc`, `memalign`, `valloc`, `posix_memalign`, every `operator new`	`__mpr_alloc`, `__mpr_calloc`, `__mpr_realloc`, `__mpr_posix_memalign` with the original arguments, a site id and a kind	allocation sites
an allocation function used as a value (stored, passed, in a table)	a thunk with the function's type that calls the runtime with its own site id	indirect calls
`memset(p, 0, n)`, `bzero`, `llvm.memset` with value 0, $n \geq 4$ KB or not constant	`__mpr_zero(p, n, site)`	zeroing
result of `vector<T>::_M_check_len` (libstdc++), `__recommend` (libc++ vector and string)	`__mpr_capacity(len, 0, sizeof(T), point)`	growth point (early)
`basic_string::_M_create(size_type&, size_type)`	the capacity argument, through `__mpr_capacity`	growth point (early)
`_Hashtable::_M_rehash(n, ...)`, libc++ `__rehash(n)`	the bucket count, through `__mpr_capacity` (8-byte buckets)	growth point (early)
a growth point inlined into the program	its id becomes the call site's, from the inlined location chain	capacity per call site
a call from the program to a container member that may grow it	bracketed by `__mpr_growth(id)` and `__mpr_growth(0)`	capacity per call site

Table 2: What `MemPrintResize` rewrites. Calls that free memory are not rewritten: `libmpr` recognises its own blocks.

Builds. For every image that an address key names, the file gets a line `# image name build-id hex`, read from the image's ELF note.

A profile can come from the same input the program will run on (we call this *own profile*, which measures the mechanism) or from other inputs of the same program (*held out*, which measures how a reservation transfers). The quantile leaves some blocks uncovered by design, and Section 9.9 shows that under-predictions are safe.

7 The `MemPrintResize` compiler pass

`MemPrintResize` is a new-pass-manager plugin for LLVM 21 [18], 574 lines of C++, loaded into clang with `-fpass-plugin`. It registers the same module pass at two extension points of the optimisation pipeline: an early instance at `PipelineStartEP`, before inlining, which handles growth points, and a late instance at `OptimizerLastEP`, after inlining, which handles everything else. It has three modes: `profile` (the default) routes allocations through the runtime, `apply` also embeds a reservation file, and `off` does nothing. Table 2 lists what it rewrites.

7.1 Allocation calls and address-taken allocators

For every call or invoke whose callee is an allocation function, the late instance builds a call to the matching runtime entry point with the original arguments, a site identifier and a kind, replaces every use of the old call, takes over its name and erases it. `operator new` under an `invoke` becomes a plain call followed by a branch to the invoke's normal destination, because the runtime reports failure the way `malloc` does, and the landing pad loses that predecessor.

An allocation function can also reach the program as a value: stored in a variable, passed to a function, or placed in a table of function pointers, as libraries that let the caller choose an allocator do. A call through such a pointer is an indirect call, and it bypasses the rewriting. The pass replaces every use of an allocation function that is not the callee of a call with a thunk, an internal function with the same type whose body calls the runtime with a site identifier of its own, derived from the location of the use. The uses in a constant, such as an initialiser, share one thunk per module. Allocations made inside shared libraries that the build does not compile reach `libmpr` through symbol interposition [33]: `libmpr` defines `malloc` and its siblings, so every library's calls resolve to it, with a return-address site and calling contexts.

7.2 Stable site identifiers

A site identifier must name the same allocation in a profiling build and an apply build, distinguish allocations that the optimiser has inlined into different callers, and survive edits elsewhere in the program. `MemPrintResize` computes a 64-bit FNV-1a hash over the callee's name, the name of the function that contains the call after inlining, and the call's debug location chain from the innermost location outward through every `inlinedAt` entry. An allocation inside an inlined `std::allocator::allocate` therefore resolves to the program line that grew the container. Each level of the chain is hashed as the file's base name, the function's name, the line relative to the first line of the function, and the column. Sample-based profiles key their counts the same way, by the offset from the function's first line, so that a profile survives edits outside the function [5, 24]. The base name keeps the identifier independent of the directory the program is built in. Without debug information the hash falls back to the function's name and the call's position in it, so the build needs `-g` (line tables are enough). Ayupov et al. match stale profiles to a new binary by the structure of its code [3]. We use the simpler line-offset scheme and detect what does not match: `scripts/check_reservations.py` lists the reservations whose identifiers a new build no longer has, from the site map that the pass writes (`0x...  kind file:line function < ...`).

7.3 Container capacities

A `std::vector` that grows by `push_back` doubles its capacity, copies its elements, and frees the old buffer, so for a moment it holds both. libstdc++ computes the new capacity in `_M_check_len`, libc++ in `__recommend`, `std::string` in `_M_create` (which takes the capacity by reference), and a hash table chooses its new bucket count before `_M_rehash`. After inlining these calls no longer exist, so the early instance wraps them before inlining:

```
%len = call i64 @_ZNKSt6vectorI...E12_M_check_lenEmPKc(ptr %this, i64 1, ptr @.str)
%cap = call i64 @__mpr_capacity(i64 %len, i64 0, i64 8, i64 <point>) ; replaces %len
```

`libmpr` returns at least `len`, so the capacity only grows: a wrong prediction cannot shrink a container, and an excess is backed lazily. Three refinements make the prediction precise.

Element size. `__mpr_capacity` receives a count of elements, and the profile measures bytes. The pass reads the element size from the debug information: the function that contains the growth point is a member of the container's class, whose first template type parameter is the element type, and its size follows from the DWARF type entry through typedefs and qualifiers [6]. A hash table's buckets are pointers (8 bytes), and when the debug information has no parameters the pass passes 1.

One capacity per call site. The growth point sits in a template instantiation, such as `_M_realloc_insert<long>`, which every vector of that type in the program shares. A single capacity for all of them would give the final size of the largest vector to every small one. When

the optimiser inlines the growth path into the program's code, as clang at `-O2` did for most `std::vector` growth in our programs, the late instance finds `__mpr_capacity` with an inlined location chain that ends at the program's call, and it replaces the instantiation's identifier with the call site's. When the growth path stays a call, for example `basic_string::_M_mutate` or `_Hashtable::_M_insert_unique_node`, the pass brackets every call from the program's own code to a container member that may grow it (`push_back`, `insert`, `append`, `operator[]` and the libraries' internal growth functions, recognised from the demangled name) with `__mpr_growth(id)` before it and `__mpr_growth(0)` after it, on both edges of an `invoke`. A function is the program's own when its debug information names a file outside the system's C++ headers. `__mpr_capacity` then uses the per-thread call site that `__mpr_growth` set. This is a calling context of depth one, maintained by the program as in probabilistic calling context [4]. The libstdc++ profile mode and Perflint report a vector's suggested initial size to the programmer [10, 23], and an IBM patent sets a Java array's initial size from the highest index accessed per allocation context [32]. Our capacity is set automatically, per call site, and only ever raised.

Small containers. A call site can grow many small containers and a few large ones. The `cap` model reserves the final size only once a container asks for more than the largest size that the call site's small containers reached in the profile.

7.4 Zeroing

The late instance replaces `memset(p, 0, n)`, `bzero` and the `llvm.memset` intrinsic with a zero value by `__mpr_zero(p, n, site)` when n is at least 4 KB or not a constant. A call that the compiler would have emitted as a `memset` library call already reaches `libmpr`'s `memset`, but code built with `_FORTIFY_SOURCE` calls `__memset_chk` instead [8], and a static binary has no symbol to interpose. Smaller constant clears stay as they are, because the compiler expands them into stores.

7.5 Embedding reservations

In apply mode the late instance reads the reservation file and adds it to the module as a constant string, `__mpr_embedded_reservations`. Every translation unit carries a copy, so the global has `weak_odr` linkage in a COMDAT group of the same name, and the linker keeps one. (`weak_any` kept 48 copies in darknet's shared library, and `linkonce_odr` was discarded as unreferenced.) `libmpr` declares the symbol weak and reads it at start-up when `MPR_RESERVATIONS` is not set, so an apply build needs no environment.

7.6 Toolchain notes

clang parses `-mllvm` options before it loads a `-fpass-plugin`, so the plugin's options are not yet registered when they are parsed. The pass therefore reads `MPR_PASS_MODE`, `MPR_PASS_RESERVATIONS` and `MPR_PASS_SITEMAP` from the environment, and the same settings remain available as options under `opt`. Our clang installation carries a configuration file that puts a newer GCC's headers first, so builds pass `-gcc-install-dir` for the system GCC 8, whose libstdc++ the programs run with, and `-no-default-config` where that newer GCC's intrinsics headers break the build. A build script (`scripts/resize_build.sh`) builds the plugin and `libmpr`, sets these variables, links with `-build-id`, and runs the program's own build command. Five `opt`-plus-FileCheck tests check the rewriting of allocation calls, growth points of the three container kinds, zeroing and thunks, and end-to-end tests build programs with the pass, profile them and run them with their reservations.

8 The `libmpr` runtime

`libmpr` is about 1,000 lines of C. It defines `malloc`, `calloc`, `realloc`, `reallocarray`, `free`, the aligned allocators, `malloc_usable_size`, the `operator new` variants, `memset`, `__memset_chk` and `bzero`, so it serves both a preloaded binary and a binary built with `MemPrintResize`. A block smaller than `MPR_MIN` (64 KB) or at a site without a reservation goes to glibc through `__libc_malloc` and its siblings, unchanged.

8.1 Finding a block's reservation

A request names its site by return address (preloaded) or by identifier (pass). When the reservation file has context lines for that site, `libmpr` takes the frames above the call with glibc's `backtrace()`, finds the call's own return address among them, and looks up the deepest context line that matches the next frames, then shallower ones, then the site's own line. `backtrace()` loads `libgcc_s` on its first use, which allocates, so `libmpr` calls it once at start-up and guards against reentry. Only sites with context lines pay for a backtrace, and only for large requests.

8.2 Serving a block

For a large block at a reserved site, `libmpr` creates one anonymous mapping of the full size with `MAP_NORESERVE` and maps the predicted part again with `MAP_FIXED` and without `MAP_NORESERVE`, which makes it accountable: the first p bytes for a prefix model, or each range of a `runs` model. The program receives a pointer to all n bytes. If it touches beyond the prediction, the kernel backs the page as for any anonymous memory, with no signal the program could observe. A design that protected the remainder with `PROT_NONE` and grew the block on a fault would also break system calls, because a `read()` into a protected page returns `EFAULT` instead of faulting. A `dead` site's block is entirely `MAP_NORESERVE`. A large `calloc` needs no `memset`, because fresh anonymous pages are zero. Blocks live in an open-addressing table keyed by address and in a sorted array of start addresses, which finds the block that holds an arbitrary address. `free` recognises a runtime block by its page alignment and a table lookup, records its resident bytes with `mincore` for telemetry, and unmaps it. The tables are themselves mapped `MAP_NORESERVE`, so `libmpr` commits nothing of its own.

8.3 Growing and shrinking

A runtime block grows without copying where possible. `libmpr` first tries to map the new tail directly after the block with `MAP_FIXED_NOREPLACE`. If the address is taken, it reserves a new range and moves the block's two mappings into it with `mremap(MREMAP_FIXED)`, which moves page tables instead of data. Growth is geometric, at least 1.5×, so a table grown one element at a time does not remap at every step, and a shrink is kept in place unless it halves the block. We found two kernel behaviours on the way. The kernel keeps a tail that was added in place and written separately as its own mapping, and `mremap` fails with `EFAULT` on a range that spans several mappings. A failed `mremap(MREMAP_FIXED)` has also already unmapped its destination. Open MPI's start-up (in `libpciaccess`) grows a table one element at a time past the size threshold and crashed on the first version. `libmpr` now maps the destination again and copies such a piece. A block committed in ranges consists of many mappings, so it grows by a new block and a copy.

8.4 Container capacities

`__mpr_capacity(len, size, elem, point)` looks up the reservation of the call site that `__mpr_growth` set on the thread (`cap:0x<site>`), else of the growth point (`id:0x<point>`, which for an inlined growth point is already the call site's). For a `cap A B` model it returns $\max(\texttt{len}, \lceil A/\texttt{elem} \rceil)$ once $\texttt{len} \cdot \texttt{elem} > B$, and `len` otherwise.

8.5 Skipping the zeroing of untouched pages

Programs that clear a buffer they have just allocated make every page of it resident. When `memset`, `__memset_chk`, `bzero` or `__mpr_zero` clears at least 16 KB inside a runtime block, wherever in the block it starts, `libmpr` reads the range's entries from `/proc/self/pagemap` [20] and writes zeros only to pages that need them:

- a page that is neither present (bit 63) nor swapped (bit 62) has never been touched since the block was mapped, and it reads as zero, so it is skipped;
- a page that is present but not mapped exclusively (bit 56 clear) is the kernel's shared zero page, mapped by an earlier read, or a page shared with a forked child, so `libmpr` reads it and skips it if it is all zeros, which is correct in both cases;
- every other page, and the partial pages at both ends, are written.

Other `memset` calls go to glibc's implementation, found with `dlsym(RTLD_NEXT)` at start-up (a byte loop that GCC must not turn back into a `memset` serves the calls made before that).

The first version checked only the present bit. llama.cpp clears its KV cache a second time after a warm-up decode, and attention had read the whole cache by then, so every page mapped the zero page and looked present. After the zero-page check, 123 MB of the cache were still resident. The cause was transparent huge pages, which this machine enables for all anonymous memory: the warm-up writes a few rows into each layer's K and V tensors (about 3 MB each), and each first write faults in a 2 MB page [22]. `libmpr` therefore marks the pages whose zeroing it skipped `MADV_NOHUGEPAGE`. Those are the pages it chose to leave untouched, and a huge page would undo the choice. Pages the program has touched keep huge pages. `MPR_THP=0` and `MPR_THP=1` disable huge pages for all runtime blocks or leave them as the system sets them, and `MPR_ZERO=0` turns the zero skipping off.

8.6 Builds

When the reservation file records an image's build identifier, `libmpr` reads the build identifier of the image it has loaded from its `PT_NOTE` segment, the first time it makes a key in that image. If the two differ, or the loaded image has none, it ignores that image's address keys and says so once on standard error. The pass's identifiers do not depend on the build and are not checked.

8.7 Telemetry and settings

With `MPR_REPORT`, `libmpr` writes per site at exit the number of blocks, the bytes requested, predicted (committed), resident at free, the blocks that used more than predicted, and the zeroing it skipped. `__mpr_committed()` returns the predicted committed total for a scheduler to read. `MPR_SCALE` multiplies every prediction for robustness tests, and `MPR_ALL` serves every large block, reserved or not, to measure the mechanism alone. Two lock-related fixes were needed: `dladdr`, used to name a return-address site, is called outside `libmpr`'s lock, and the exit handler releases the lock before it writes the report, because `fclose` frees its buffer through `free`.

9 Evaluation

9.1 Setup

All runs used a two-socket Intel Xeon E5-2620 v4 (16 cores, 32 threads) with 62 GB of RAM, Linux 4.18 (RHEL 8) with heuristic overcommit and transparent huge pages set to `always`, GCC 8.5 and clang 21. Each configuration ran three times, and we report the mean of the peak RSS and the peak commit, sampled every 20 ms for the program and its children. Outputs were compared by hash: all printed lines except those that report a time or a rate, the modularity line for miniVite, the predictions for darknet, and the generated text for the LLM engines.

The workloads are:

- darknet AlexNet inference on 4 images [37], built with GCC (preloaded) and with `MemPrintResize`;
- miniVite [12] with 32768 vertices and 4 OpenMP threads, built with GCC (preloaded) and with `MemPrintResize`;
- llama.cpp b11500 [11], CPU only, 4 threads, greedy decoding of "Once upon a time" with SmolLM2-135M-Instruct (32 and 128 tokens) and SmolLM2-360M-Instruct (64 tokens), both quantised to Q8_0 [1], at the model's default 8192-token context, built with GCC (preloaded) and with clang and `MemPrintResize`;
- llama2.c [15], greedy, 128 steps, with the stories15M, 42M and 110M models;
- the 23 programs of the scheduling corpus [30] at their largest inputs: PolyBench kernels, GAP, HPCCG, LULESH, miniFE, Kripke, Quicksilver, NPB, darknet, miniVite, the Lua, Perl and Python interpreters, SQLite and GCC's cc1.

Each input has an *own* profile and a *held-out* profile from the program's other inputs (for darknet, the reference and Extraction networks; for miniVite, 8192 and 16384 vertices). The profiles use the Pin tool with `-sites`, which includes calling contexts and `-skip_zeroing`. We call the system as first built (prefix models only, one site per call, the first zero-skipping runtime) the *first version*, and the system with calling contexts, shaped reservations, zeroing left out of the profile, and per-call-site container capacities the *refined* version.

9.2 Headline results

Table 3 and Figure 2 summarise the results. With its own profile, `libmpr` reduces peak commit by 86.3% on llama.cpp with the 135M model (404.5 to 55.6 MB) and by 90.0% with the 360M model (560.5 to 56.3 MB), and it reduces peak RSS by 53.9% and 45.5%. It reduces commit by 84.5% on llama2.c 110M (78.2 to 12.1 MB) and by 46.8% on darknet. Held-out profiles do as well on llama.cpp, where the sites are the same code paths at every model size. Every run of every workload produced glibc's output, and run time stayed between 0.93× and 1.02× of glibc.

9.3 Effect of the refinements

Figure 3 and Table 4 compare the first and the refined version. The refinements reduce llama.cpp's commit by a further 79% (265.1 to 55.6 MB) and llama2.c 110M's by 83% (72.8 to 12.1 MB). Three refinements combine on llama.cpp:

Workload	glibc RSS	glibc Commit	libmpr, own profile RSS	libmpr, own profile Commit	libmpr, held out RSS	libmpr, held out Commit	Output identical
darknet AlexNet	287.3	534.7	287.7	284.6	287.8	519.1	yes
miniVite 32768 (pass)	30.3	46.2	27.0	26.5	28.6	32.2	yes
llama.cpp 135M, 32 tok	380.0	404.5	175.1	55.6	175.2	56.6	yes
llama.cpp 135M, 128 tok	380.1	404.5	177.4	56.4	177.5	56.2	yes
llama.cpp 360M, 64 tok	768.2	560.5	419.0	56.3	424.4	57.4	yes
llama2.c 15M	67.0	6.2	67.4	4.3	67.4	3.3	yes
llama2.c 42M	201.0	36.2	175.9	6.8	175.8	21.6	yes
llama2.c 110M	514.6	78.2	452.5	12.1	469.6	45.8	yes

Table 3: Peak RSS and peak commit in MB (mean of 3 runs) with the refined system. miniVite's glibc column is the GCC build; its libmpr columns are the `MemPrintResize` build. All other libmpr columns are `libmpr` preloaded into the GCC build.

Workload	Peak commit (MB) glibc	Peak commit (MB) first	Peak commit (MB) refined	Peak RSS (MB) glibc	Peak RSS (MB) first	Peak RSS (MB) refined
llama.cpp 135M, 32 tok	404.5	265.1	55.6	380.0	193.2	175.1
llama.cpp 360M, 64 tok	560.5	419.9	56.3	768.2	435.4	419.0
llama2.c 42M	36.2	32.8	6.8	201.0	197.6	175.9
llama2.c 110M	78.2	72.8	12.1	514.6	486.8	452.5
miniVite 32768 (pass)	34.0	26.1	26.5	30.3	27.0	27.0

Table 4: The numbers of Figure 3 (MB, mean of 3 runs).

- **Zeroing left out of the profile.** The first profile counted the start-up `memset` as touching the whole 189 MB KV cache, so the cache was committed in full. With `-skip_zeroing`, the profile sees the rows the model writes.
- **Calling contexts.** The KV cache, the compute buffer and several small buffers are allocated through one site in `ggml_aligned_malloc`. The fit splits that site by two caller frames, and each buffer gets its own model.
- **Shaped reservations.** The KV cache is 60 tensors (30 layers, K and V), each written in its first rows. Its `runs` reservation commits 1.0 MB of the cache, against 189 MB for any prefix, and the compute buffer's commits 0.4 MB of 115 MB.

On llama2.c, the shaped reservation alone does the work: each KV cache gets one committed range per layer. RSS falls further too (llama2.c 110M: 486.8 to 452.5 MB), because ranges committed as separate mappings no longer form transparent huge pages across the untouched rows between them. On miniVite, the refined version matches the first version's hand-made capacity, with capacities fitted automatically per call site (Section 9.5).

9.4 darknet: dead training buffers

Figure 4 shows what the profile found. darknet's largest site allocates the `weight_updates` buffers of its connected layers (`connected_layer.c:35`) and convolutional layers (`convolutional_layer.c:196`), which only training uses, so inference never touches them. With its own profile, `libmpr` maps them `MAP_NORESERVE`, and peak commit falls from 534.7 MB to 284.6 MB, close to

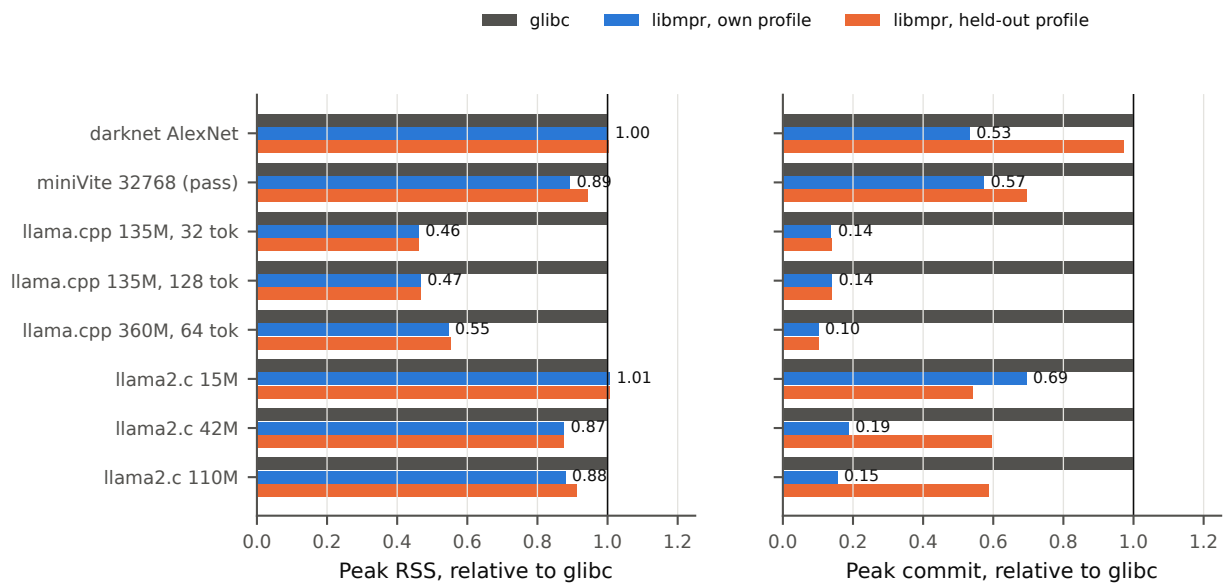


Figure 2: Peak RSS and peak commit under `libmpr`, relative to glibc. Every bar is a run whose output was identical to glibc's.

the 270 MB the program touches. RSS does not change, because the untouched buffers were never resident. The `MemPrintResize` apply build, with the reservations embedded and no environment, gives the same 284.6 MB. A held-out profile from the reference and Extraction networks saves only 15.6 MB, because those networks have no fully connected layers: the profile has no data about the connected layers' site, which holds 234 MB of AlexNet's untouched buffers, and only the convolutional layers' 15 MB carry over. A profile helps the code paths it covers, and no refinement changes that.

9.5 miniVite: container capacities and where its memory goes

miniVite touches 15.4 MB at its peak but holds 28.7 MB of RSS at its peak RSS and commits 44.0 MB at its peak commit (Figure 5, left). The breakdown, from `/proc/<pid>/smaps` at each peak, explains the gap:

- **Code, not data, is a third of the RSS.** 9.6 MB is library and program code (Open MPI, libstdc++, libgomp), which MemPrint does not count because it measures data references, not instruction fetches.
- **Vector growth.** `edgeList` (`graph.hpp:777`) grows by `emplace_back` without `reserve`, and it is the only large heap site live at the peak. When it doubles, the old 6.3 MB buffer and the new 12.6 MB buffer are both live (18.9 MB requested, 12.8 MB touched), and huge pages back the new buffer's 6.5 touched MB with about 8 MB.
- **Small blocks pay glibc's per-chunk overhead.** 1.8 million blocks smaller than 64 KB (hash table nodes of 24 to 47 bytes and small vectors) request 57 MB over the run but occupy 81 MB of glibc chunks (1.42×), and the heap holds them in 10.4 MB at the peak.
- **Untouched library memory.** Open MPI's start-up allocates 1.6 MB that it never touches.

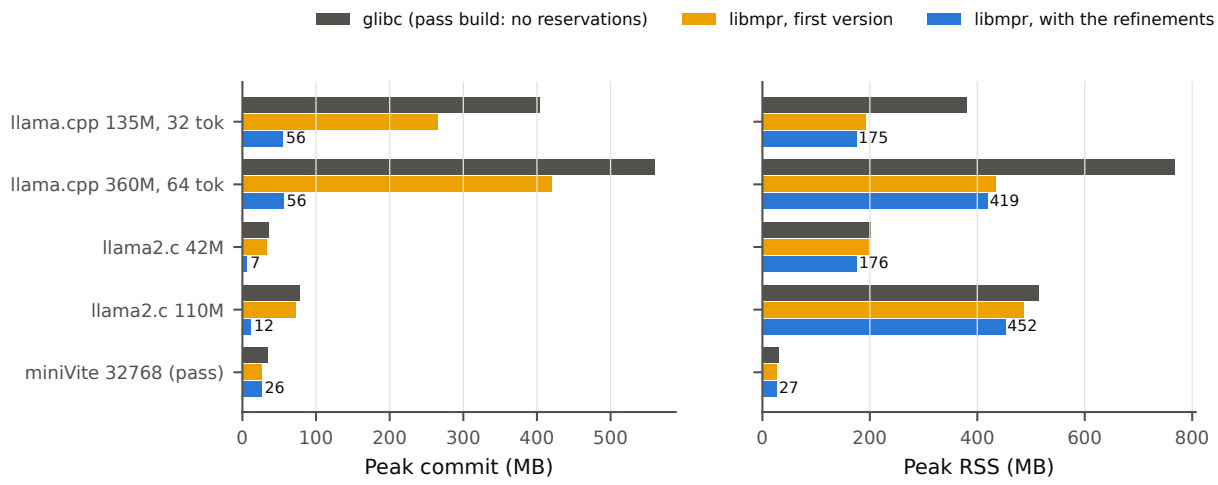


Figure 3: Peak commit and peak RSS under glibc, the first version of the system and the refined version, with own profiles. miniVite is the `MemPrintResize` build; its glibc bar is the build without reservations.

- **Commit is mostly thread stacks.** 32 MB of the peak commit is four 8 MB OpenMP thread stacks, of which 2 MB is resident. A smaller `OMP_STACKSIZE` would remove that commit, which no allocator change can.

Built with `MemPrintResize`, miniVite has 20 growth points, 23 of them inlined into call sites in its own code, and 44 calls to container members bracketed as growth contexts. The profile of the 32768-vertex input yields 12 capacities: `edgeList`'s final size (6.5 MB, one allocation instead of eight totalling 25 MB) and those of 11 vectors whose growth stays out of line, reached through growth contexts. Two call sites, a vector and a hash table, each grow about half a million small containers and get no capacity, because their chains never reach 64 KB. With its own profile, peak RSS falls from 30.3 MB to 27.0 MB (11%) and peak commit from 34.0 MB to 26.5 MB (22%), with the same modularity and run time (1.79 s against 1.80 s). The held-out profile from 8192 and 16384 vertices reserves the 16384-vertex final size, half of what the 32768-vertex edge list needs, and saves less (28.6 MB RSS, 32.2 MB commit).

The first measurement of the refined pass ran 15% slower with reservations. The cause was `libmpr`'s capacity lookup, which formatted a key, took a lock and searched the site list at every growth of every container, about a million times in this run. A table by numeric identifier, filled when the reservations load and read without the lock, removed the cost, and the build without reservations became 17% faster as well (2.17 to 1.80 s).

9.6 LLM inference

llama.cpp. With the 135M model, llama.cpp allocates three large buffers through `ggml_aligned_malloc`: a 188.7 MB KV cache for 8192 tokens, a 115 MB compute buffer of which 0.3 MB is touched, and two 44 MB scheduler arrays (`ggml_backend_sched_new`) that are never touched. It clears the whole KV cache with `memset` at start-up and again after its warm-up decode. Under glibc, the whole cache is therefore resident although a 32-token completion uses 0.4% of it. The model's weights are mapped from the GGUF file, so their 143 MB count in RSS but not in commit.

RSS and commit fall for different reasons (Figure 6). RSS falls from 380 MB to 175 MB because of the zero skipping. To attribute it, we recorded each buffer's address and measured its

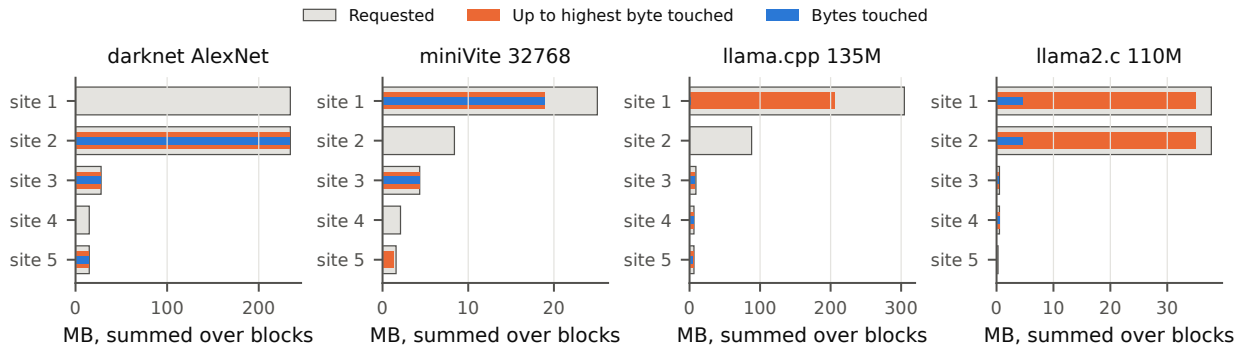


Figure 4: The five largest allocation sites of four workloads, summed over their blocks: bytes requested, bytes up to the highest touched offset (what a prefix reservation must commit), and bytes touched, from the refined profiles. darknet's first site is dead. llama.cpp's first site is `ggml_aligned_malloc`, which serves the KV cache and the compute buffer; with zeroing left out, few of its bytes count as touched. llama2.c's two KV caches are touched sparsely across the whole block.

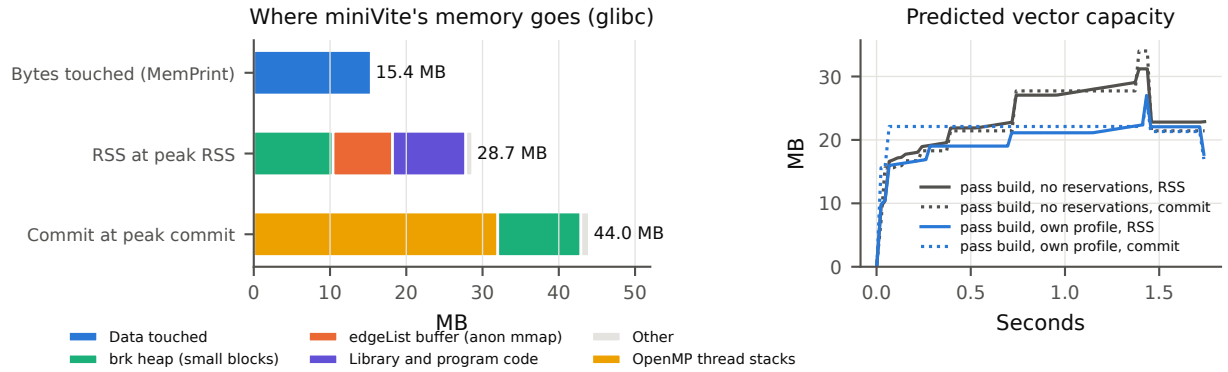


Figure 5: Left: miniVite's bytes touched, its RSS at peak RSS and its commit at peak commit under glibc, by mapping (smaps, 32768 vertices, 4 threads). Right: RSS and commit over time of the `MemPrintResize` build without reservations and with its own profile.

resident pages at the peak from `/proc/<pid>/pagemap` under four configurations of the first runtime. Without zero skipping, the KV cache holds 188.7 MB, as under glibc. Skipping the zeroing alone leaves 123.7 MB, all of it huge pages faulted in by the warm-up's writes of one row per layer. Skipping the zeroing and keeping huge pages off the skipped pages leaves 1.0 MB. The compute buffer holds 15 to 17 MB under every configuration, which is the part llama.cpp uses at the peak, and the remaining 174 MB (weights, heap and code) does not change. Commit falls from 404.5 MB to 55.6 MB because of the reservations (Section 9.3): the scheduler arrays are `dead`, and the KV cache and the compute buffer are committed in ranges. Run time does not change (0.71 s against 0.73 s for 32 tokens, 1.81 s against 1.95 s for the 360M model).

llama2.c. llama2.c's KV caches are also sized for the full sequence (37.7 MB each for the 110M model, of which 4.7 MB is touched for 128 of 1024 positions). It allocates them with `calloc` and does not clear them, so their untouched part was never resident under glibc either. They are laid out as [layer][position][dimension], so every layer's first rows are touched and the highest touched offset lies at 93% of the block (Figure 4). A prefix reservation must cover it: the first version committed 72.8 of 78.2 MB. The shaped reservation commits one range per layer, 12.1 MB in all.

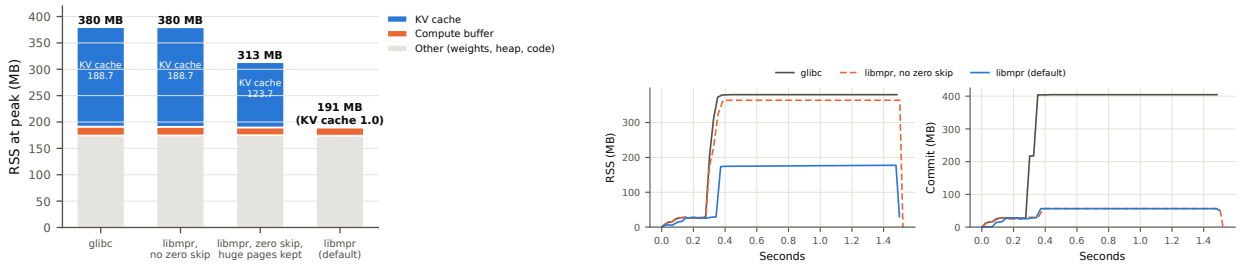


Figure 6: llama.cpp with SmolLM2-135M. Left: RSS at the peak by buffer, measured from `/proc/<pid>/pagemap` over each buffer's address range (32 tokens, first version of the runtime; "other" is the mapped weights, the heap and code). Right: RSS and commit over a 128-token run with the refined system.

	Preloaded (GCC build)			Built with `MemPrintResize` (clang)		
Input	none	own	held out	none	own	held out
135M, 32 tokens, commit	404.5	55.6	56.6	405.0	56.0	57.1
135M, 32 tokens, RSS	380.0	175.1	175.2	382.4	177.5	177.7
135M, 128 tokens, commit	404.5	56.4	56.2	405.0	56.9	56.6
135M, 128 tokens, RSS	380.1	177.4	177.5	381.8	179.6	179.9
360M, 64 tokens, commit	560.5	56.3	57.4	560.9	56.9	57.9
360M, 64 tokens, RSS	768.2	419.0	424.4	770.0	421.5	427.6

Table 5: llama.cpp, peak commit and peak RSS in MB: the GCC build with `libmpr` preloaded (sites are return addresses) against the clang build with `MemPrintResize` (sites are the pass's identifiers), each with no reservations, its own profile and held-out profiles.

The held-out profile, from models with 6 and 8 layers, places its ranges where a 12-layer model's are only in part, and commits 45.8 MB.

9.7 Built with the pass, or preloaded

Table 5 compares the two ways of running llama.cpp. They save the same: with its own profile the clang build with `MemPrintResize` commits 56.0 MB and holds 177.5 MB (135M, 32 tokens), against 55.6 MB and 175.1 MB for the preloaded GCC build, and held-out profiles give the same in both. In llama.cpp the large buffers come from a few allocation functions, which return addresses name as well as the pass's identifiers do, and none of its growing containers reaches 64 KB of capacity in a way that a reservation changes. The pass is needed for container capacities, which come from inside inlined library code, and it gives identifiers that do not depend on the build; a preloaded run needs no rebuild. Table 6 shows what the pass costs: no measurable compile time on darknet and 2.8% on llama.cpp (one build, while other jobs ran). With the previous identifiers, inserting lines at the top of five darknet files changed every identifier in those files (80.9% of all sites unchanged); with line offsets and base file names, only the allocations below a line inserted inside a function change (96.6% unchanged). Rebuilding the program changes its build identifier, and `libmpr` then refuses the profile's address keys, as it did for llama.cpp's calling contexts after a rebuild: the KV cache's context lines no longer applied, and its commit rose to 261 MB until the program was profiled again.

Measure	darknet	llama.cpp
Allocation sites rewritten	631	5,137
Zeroing sites, growth points, inlined capacity call sites, growth contexts	–	720, 442, 2,315, 2,066
Build CPU time without / with the pass (s)	24.59 / 24.58	738.2 / 758.6
Site ids unchanged after inserting lines (previous pass / `MemPrintResize`)	80.9% / 96.6%	–

Table 6: What the pass costs and how stable its identifiers are. darknet: mean of three builds; llama.cpp: one build each, while other jobs ran. The id test inserts three lines at the top of five darknet files and one line inside `make_connected_layer`; with `MemPrintResize`, the 22 identifiers that change are all allocations in that function below the inserted line.

23 programs, largest inputs	glibc	libmpr, own	libmpr, held out
Total peak commit (MB)	2,504	2,236	2,482
Total peak RSS (MB)	2,290	2,292	2,291
Median commit / touched	1.01	1.01	1.01
Median RSS / touched	1.05	1.06	1.06
Time relative to glibc (range)	1	0.95 to 1.07	0.86 to 1.07
Programs with identical output	–	23 of 23	23 of 23

Table 7: The 23 programs of the scheduling corpus at their largest inputs (3 runs each), with the refined system.

9.8 The corpus: no regressions

Table 7 and Figure 7 show the 23-program corpus, profiled again with the refined Pin tool. Under glibc, the median program commits 1.01× its bytes touched and holds 1.05× in RSS, so there is little for a resizer to take. The exceptions are darknet (1.98× commit), miniVite (3.00×), Quicksilver (1.25×), cc1 and Python (1.13×) and Lua (1.12×). With own profiles, total peak commit falls by 10.7%, from 2,504 MB to 2,236 MB, most of it darknet's, and Perl's commit falls below its bytes touched (0.92×). Held-out profiles save 0.9%. Total RSS does not change (2,290 to 2,292 MB). Every program produced glibc's output in every run, and run time stayed between 0.95× and 1.07× of glibc with own profiles. One regression appeared while we built the refined runtime and was fixed: `libmpr`'s site table, larger once each site can hold 128 ranges, was backed by a 2 MB huge page at its first entry, which added 2.2 MB of RSS to every program until the table was marked `MADV_NOHUGEPAGE`.

9.9 Robustness to wrong predictions

Under-prediction is safe by construction, and we tested it. With every prediction multiplied by 0.1 (`MPR_SCALE=0.1`), darknet and miniVite produced identical outputs, and the telemetry reported the blocks that used more than predicted. The quantile in the fit leaves blocks uncovered by design, and held-out profiles under-predict on the LLM engines and the corpus, again with identical outputs. Over-prediction costs commit and nothing else, because the committed part is still backed lazily. A container capacity can only raise the capacity a container would have chosen. Two control programs with no waste, PolyBench 2mm and HPCCG, with `libmpr` serving every large block (`MPR_ALL`), changed by at most 1.8 MB of RSS and 0.1 MB of commit.

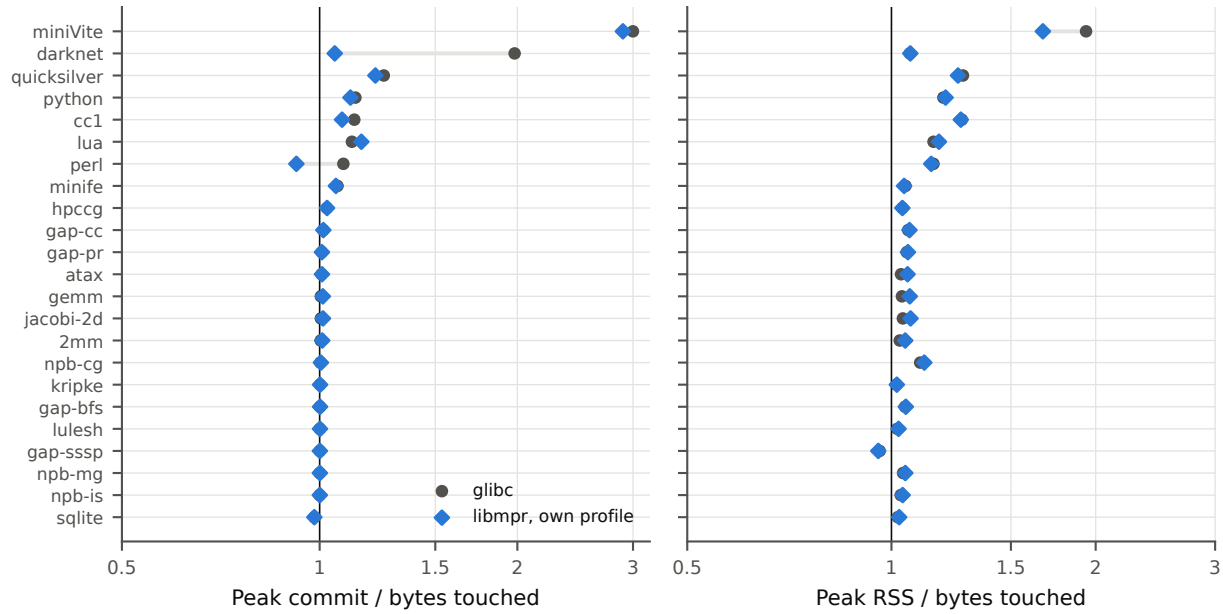


Figure 7: Peak commit and peak RSS relative to bytes touched for the 23 corpus programs, under glibc and under `libmpr` with each program's own profile. Most programs already sit at about 1.

Oracle	Truth: bytes touched	Truth: RSS	Truth: RSS under libmpr
Exact curve	90	91	103
Exact peak	460	436	457
MemPrint W5	391	1,807	1,727
DHAT peak	979	1,277	1,230
User request	2,215	2,197	2,204

Table 8: Mean response time (minutes) at 80% offered load in the DMSched-Sim scheduler of our earlier study [30], with calibrated margins, under three truths of what a job holds. The third truth is each program's RSS when it runs with `libmpr` (first version) and its own profile.

9.10 Effect on scheduling

Our earlier study replayed the corpus's measured footprints through a simulator of 32 nodes with a shared memory pool [30]. We added a third truth, the RSS of each program under `libmpr` (Table 8). The results hardly move from the RSS truth, because the corpus programs' resident memory hardly moved: the exact curve gives 103 minutes (91 under the RSS truth) and MemPrint's windowed curves 1,727 (1,807). The refined system leaves the corpus's total RSS unchanged as well. The scheduling study's conclusions therefore do not depend on resizing allocations, since its gain comes from reserving a curve instead of a peak. Resizing matters for programs that commit or hold much more than they touch, such as darknet and the LLM engines, and for systems with strict overcommit, where the saved commit is memory that can be promised to other jobs.

MemPrint curve	Median margin	90th percentile	Response (min)
Full trace, bytes, references	0.06	0.58	1,847
Full trace, bytes, seconds	0.08	0.44	1,476
Full trace, pages, references	0.03	0.56	1,942
Full trace, pages, seconds	0.05	0.41	1,423
W5, bytes, references	0.04	0.58	1,807
W5, bytes, seconds	0.04	0.56	1,820
W5, pages, references	0.04	0.58	1,979
W5, pages, seconds	0.04	0.49	1,705
Exact RSS peak	–	–	436
Exact RSS curve	–	–	91
User request	–	–	2,197

Table 9: MemPrint's footprint curves under the RSS truth, with the Pin tool's clock and unit options: the margin (headroom as a share of the curve's peak) each program needs for the curve never to fall below its RSS, over 23 programs, and the mean response time at 80% load with calibrated margins.

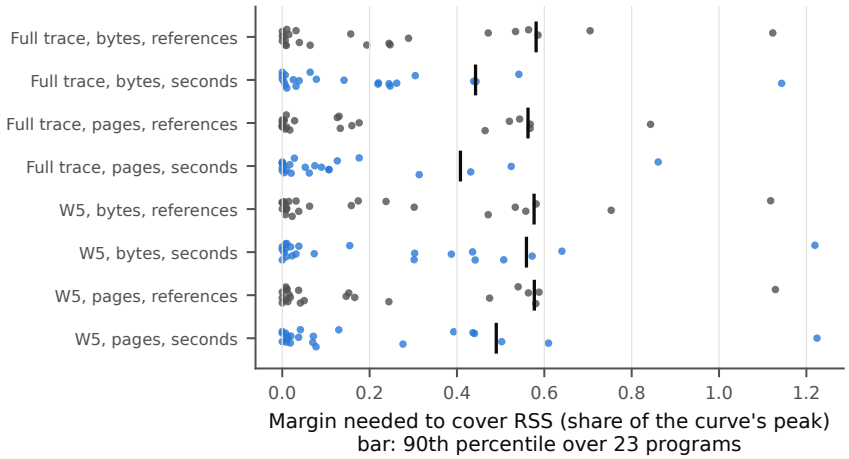


Figure 8: The margin each corpus program needs (dots; blue: wall-clock axis) and the 90th percentile over programs (bars), which sets the calibrated margin.

10 Discussion

10.1 The time axis of a footprint curve

In the scheduling study, a MemPrint curve needed about 60% headroom to cover the RSS truth, and we attributed part of that to the time axis, since MemPrint counts memory references while RSS is sampled in seconds [30]. The new options let us test the attribution. We reran the full trace and the 5% windowed mode at every program's largest input with `-footprint pages`, and mapped each curve to the wall time recorded in the `Seconds` column (Table 9, Figure 8). On wall time, the full trace's 90th-percentile margin falls from 0.58 to 0.44, and the mean response time from 1,847 to 1,476 minutes. The programs that gain most have phases that are short in references but long in time, such as GAP BFS (margin 0.53 to 0.06), GAP CC (0.56 to 0.09) and Lua (0.70 to 0.05). Counting pages instead of bytes helps the programs whose touched bytes are scattered (cc1 0.25 to 0.13, Perl 0.29 to 0.13, miniVite 1.12 to 0.84), but it does not reduce the calibrated margin on its own. Together the two options give 0.41 and 1,423 minutes, still far from the exact

peak's 436 minutes. The remaining margin comes from memory that MemPrint does not count, such as miniVite's library code, which the scheduler's RSS baseline (the run's smallest sample, taken before MPI loads its libraries) treats as footprint. The wall time is measured under Pin, which slows phases unevenly, so it approximates native time.

10.2 Bytes touched can exceed the resident set

With the first profile, llama.cpp's RSS under `libmpr` (193 MB) fell below its bytes touched as MemPrint measured them (353 MB at the peak, with the mapped weights). The difference was the zeroing `memset`: writing zeros to fresh anonymous memory counts as touching it, although no reader can distinguish the result from untouched memory. `-skip_zeroing` now leaves such writes out of the profile (Section 5), which is what let `libmpr` commit 1 MB of the KV cache instead of 189 MB. It is on with `-sites` only: the footprint truth of the scheduling study keeps its definition, in which a write is a touch whatever it writes. Whether the truth should also leave out zeroing of untouched memory is a question of definition. Under it, the truth would follow the resident set of a program run with `libmpr` more closely, and the resident set of a program run without it less closely.

10.3 What the RSS numbers measure

The RSS figures in this paper are peaks of samples of `smaps_rollup`, taken every 20 ms, so a spike shorter than the interval can be missed. They include file-backed pages and count a huge page as 2 MB. The per-buffer figures for llama.cpp come from a different instrument. A logging library recorded where glibc (or `libmpr`) placed each buffer of at least 1 MB, without changing how it allocated, and at the sample with the largest total RSS we counted each buffer's pages that pagemap reports as present and mapped exclusively, which excludes the shared zero page. Under glibc the KV cache and the compute buffer lie in one merged mapping, so `smaps` alone cannot separate them.

10.4 Where resizing pays

The results separate three cases. A program that allocates buffers it never touches (darknet, llama.cpp's scheduler arrays) gains commit but no RSS, because the kernel never backed those buffers. A program whose allocator or own code touches memory it does not need (a zeroing `memset`, vector doubling, huge pages over sparse writes) gains RSS, and only a change in how the memory is obtained recovers it. A program that already uses what it allocates, which is the median program in our corpus, gains nothing and loses nothing.

11 Limitations and future work

A limitation of the container capacities is that a capacity measured on one input transfers to another only as far as the final sizes agree. The final size of miniVite's edge list depends on the graph, so a capacity from the 8192- and 16384-vertex inputs reserves the 16384-vertex size, half of what the 32768-vertex input needs, and the vector still grows once more (Section 9.5). Predicting the final size from the input would need a model of size against the input's parameters, or MemPrint's static route, which estimates footprints from source and does not yet produce per-site profiles. The growth points are specific to the containers of libstdc++ and libc++ listed in Table 2; `std::deque` grows in fixed 512-byte nodes and has no capacity to raise.

A limitation of the shaped reservations is that a range is a fraction of the block, so it transfers between blocks of the same layout. llama2.c's held-out reservations, from models with 6 and 8 layers, cover a 12-layer model's KV caches only partly, and commit 46 MB against 12 MB with the model's own profile. A reservation that names the layout (the number of periods and the used prefix of each) would transfer between models; the profile would have to find the period.

Calling contexts are taken with a backtrace for each large allocation at a site that has context lines. The frames are return addresses, so they are tied to a build, as image keys are, and a context deeper than two frames is not tried.

Small blocks are left to glibc, so the per-chunk overhead of small allocations, 1.42× in miniVite, is not addressed. Allocators inside interpreters (pymalloc, Lua's tables) and custom `operator new` are not covered beyond the large blocks they request from the system.

The commit saving depends on the overcommit policy. Under strict overcommit (`vm.overcommit_memory=2`) the kernel ignores `MAP_NORESERVE`, so the saving disappears, although correctness is unaffected; we could not set the policy on our machine to measure it. The zero skipping reads pagemap, which unprivileged processes may read on this kernel without page frame numbers, and its zero-page check treats a page shared after `fork` conservatively by reading it; no workload we ran forks. Keeping huge pages off the skipped pages cost no time on llama.cpp, but a program that clears a buffer and then fills all of it densely would run on 4 KB pages there.

Profiling is expensive: the Pin tool's exact mode with `-sites` took 300 to 1,100 s for llama.cpp runs of 0.8 to 2 s, and 617 s for llama2.c 110M (18 s natively), so profiles must be reused across runs, which the held-out results show is possible when the inputs exercise the same code paths. Whether a sampled or windowed `-sites` profile would give reservations as good is open. We ran single short completions; an LLM server that runs many sequences fills its KV cache over time, and its savings would shrink as it does.

12 Conclusion

We passed MemPrint's measured per-site footprints to a compiler pass and a runtime as non-binding reservations, so that a program is backed by what it touches rather than what it requests, and an under-prediction costs accounting instead of correctness. Every program we ran produced its original output. Calling contexts, shaped reservations and a profile that does not count zeroing as use took the system from a 34% to an 86% commit reduction on llama.cpp and from 7% to 85% on llama2.c, and zero skipping halves llama.cpp's resident memory. Per-call-site container capacities, with element sizes from the debug information, remove miniVite's vector growth without slowing it, once the runtime's lookup is fast enough for a million growths. The 23-program corpus, which already holds about what it touches, saw no regressions. The measurements also showed that a footprint curve is most useful to a scheduler on a wall-clock axis, and that what a profile counts as use, such as zeros written to fresh memory, decides what can be saved.

References

[1] Loubna Ben Allal et al. SmolLM2: When smol goes big – data-centric training of a small language model. arXiv:2502.02737, 2025.

[2] Glenn Ammons, Thomas Ball, and James R. Larus. Exploiting hardware performance counters with flow and context sensitive profiling. In *Proceedings of the ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*, pages 85–96, 1997.

[3] Amir Ayupov, Maksim Panchenko, and Sergey Pupyrev. Stale profile matching. In *Proceedings of the 33rd ACM SIGPLAN International Conference on Compiler Construction (CC)*, pages 162–173, 2024.

[4] Michael D. Bond and Kathryn S. McKinley. Probabilistic calling context. In *Proceedings of the 22nd ACM SIGPLAN Conference on Object-Oriented Programming Systems, Languages and Applications (OOPSLA)*, pages 97–112, 2007.

[5] Dehao Chen, David Xinliang Li, and Tipp Moseley. AutoFDO: Automatic feedback-directed optimization for warehouse-scale applications. In *Proceedings of the International Symposium on Code Generation and Optimization (CGO)*, pages 12–23, 2016.

[6] DWARF Debugging Information Format Committee. *DWARF Debugging Information Format, Version 5*, 2017.

[7] Jason Evans. A scalable concurrent malloc(3) implementation for FreeBSD. In *BSDCan*, 2006.

[8] Free Software Foundation. *The GNU C Library Reference Manual: Source Fortification.* `https://sourceware.org/glibc/manual/`.

[9] Free Software Foundation. *The GNU Linker (ld):* `-build-id`. GNU Binutils documentation.

[10] GCC. *The GNU C++ Library Manual, Chapter 19: Profile Mode.* `https://gcc.gnu.org/onlinedocs/gcc-4.7.1/libstdc++/manual/manual/bk01pt03ch19s07.html`.

[11] Georgi Gerganov and ggml contributors. llama.cpp: LLM inference in C/C++. `https://github.com/ggml-org/llama.cpp`, release b11500, 2026.

[12] Sayan Ghosh, Mahantesh Halappanavar, Antonino Tumeo, Ananth Kalyanaraman, and Assefaw H. Gebremedhin. miniVite: A graph analytics benchmarking tool for massively parallel systems. In *Proceedings of the IEEE/ACM Performance Modeling, Benchmarking and Simulation of High Performance Computer Systems Workshop (PMBS)*, 2018.

[13] A. H. Hunter, Chris Kennelly, Paul Turner, Darryl Gove, Tipp Moseley, and Parthasarathy Ranganathan. Beyond malloc efficiency to fleet efficiency: A hugepage-aware memory allocator. In *Proceedings of the 15th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, pages 257–273, 2021.

[14] Changhee Jung, Silvius Rus, Brian P. Railing, Nathan Clark, and Santosh Pande. Brainy: Effective selection of data structures. In *Proceedings of the 32nd ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*, pages 86–97, 2011. Also in ACM SIGPLAN Notices 46(6).

[15] Andrej Karpathy. llama2.c: Inference Llama 2 in one file of pure C. `https://github.com/karpathy/llama2.c`, 2023.

[16] Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP)*, 2023.

[17] Youngjin Kwon, Hangchen Yu, Simon Peter, Christopher J. Rossbach, and Emmett Witchel. Coordinated and efficient huge page management with Ingens. In *Proceedings of the 12th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, pages 705–721, 2016.

[18] Chris Lattner and Vikram Adve. LLVM: A compilation framework for lifelong program analysis and transformation. In *Proceedings of the International Symposium on Code Generation and Optimization (CGO)*, 2004.

[19] Huaicheng Li et al. Pond: CXL-based memory pooling systems for cloud platforms. In *International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 2023.

[20] The Linux Kernel Documentation. *Examining Process Page Tables*. Documentation/admin-guide/mm/pagemap.rst.

[21] The Linux Kernel Documentation. *Overcommit Accounting*. Documentation/mm/overcommit-accounting.rst.

[22] The Linux Kernel Documentation. *Transparent Hugepage Support*. Documentation/admin-guide/mm/transhuge.rst.

[23] Lixia Liu and Silvius Rus. Perflint: A context sensitive performance advisor for C++ programs. In *Proceedings of the 7th IEEE/ACM International Symposium on Code Generation and Optimization (CGO)*, pages 265–274, 2009.

[24] The LLVM Project. *Clang Compiler User's Manual: Sample Profile Text Format*. `https://clang.llvm.org/docs/UsersManual.html`.

[25] The LLVM Project. *MemProf: Memory Profiling for LLVM*, 2025. `https://llvm.org/docs/MemProf.html`.

[26] Chi-Keung Luk, Robert Cohn, Robert Muth, Harish Patil, Artur Klauser, Geoff Lowney, Steven Wallace, Vijay Janapa Reddi, and Kim Hazelwood. Pin: building customized program analysis tools with dynamic instrumentation. In *Proceedings of the 2005 ACM SIGPLAN Conference on Programming Language Design and Implementation*, PLDI '05, page 190–200, New York, NY, USA, 2005. Association for Computing Machinery.

[27] Martin Maas, David G. Andersen, Michael Isard, Mohammad Mahdi Javanmard, Kathryn S. McKinley, and Colin Raffel. Learning-based memory allocation for C++ server workloads. In *Proceedings of the 25th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 2020.

[28] Lukas Makor, Sebastian Kloibhofer, Peter Hofer, David Leopoldseder, and Hanspeter Mössenböck. Automated profile-guided replacement of data structures to reduce memory allocation. *The Art, Science, and Engineering of Programming*, 10(1), 2025.

[29] Microsoft. *VirtualAlloc function (memoryapi.h)*. Win32 API documentation, `https://learn.microsoft.com/windows/win32/api/memoryapi/nf-memoryapi-virtualalloc`.

[30] Nafis Mustakin. What is an exact memory footprint worth to a scheduler? measured footprints and job scheduling on disaggregated memory. MemPrint repository, paper_sched, 2026.

[31] Nafis Mustakin, Joseph Manzano, and Daniel Wong. MemPrint: Constructing program memory footprint estimations using statistical methods from sparsely sampled Pin-based memory traces. Manuscript, 2025.

[32] Rei Odaira. Information processing apparatus and program and method for adjusting initial array size. U.S. Patent 9,218,191 B2, assigned to International Business Machines Corp., 2015.

[33] Oracle. *Linker and Libraries Guide: Using Interposition.* Oracle Solaris 11.4 documentation.

[34] Ashish Panwar, Sorav Bansal, and K. Gopinath. HawkEye: Efficient fine-grained OS support for huge pages. In *Proceedings of the 24th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 2019.

[35] Bobby Powers, David Tench, Emery D. Berger, and Andrew McGregor. Mesh: Compacting memory management for C/C++ applications. In *Proceedings of the 40th ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*, 2019.

[36] Ramya Prabhu, Ajay Nayak, Jayashree Mohan, Ramachandran Ramjee, and Ashish Panwar. vAttention: Dynamic memory management for serving LLMs without PagedAttention. In *Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 2025.

[37] Joseph Redmon. Darknet: Open source neural networks in C. `https://pjreddie.com/darknet/`, 2016.

[38] Niklas Röjemo and Colin Runciman. Lag, drag, void and use: Heap profiling and space-efficient compilation revisited. In *Proceedings of the ACM SIGPLAN International Conference on Functional Programming (ICFP)*, 1996.

[39] Joe Savage and Timothy M. Jones. HALO: Post-link heap-layout optimisation. In *Proceedings of the 18th ACM/IEEE International Symposium on Code Generation and Optimization (CGO)*, 2020.

[40] Ohad Shacham, Martin Vechev, and Eran Yahav. Chameleon: Adaptive selection of collections. In *Proceedings of the 30th ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*, 2009.

[41] Valgrind Developers. *DHAT: a dynamic heap analysis tool.* Available at `https://valgrind.org/docs/manual/dh-manual.html`.

[42] Guoqing Xu, Nick Mitchell, Matthew Arnold, Atanas Rountev, Edith Schonberg, and Gary Sevitsky. Finding low-utility data structures. In *Proceedings of the 31st ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*, pages 174–186, 2010.

[43] Guoqing Xu and Atanas Rountev. Detecting inefficiently-used containers to avoid bloat. In *Proceedings of the 31st ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*, pages 160–173, 2010.

[44] Xi Yang, Stephen M. Blackburn, Daniel Frampton, Jennifer B. Sartor, and Kathryn S. McKinley. Why nothing matters: The impact of zeroing. In *Proceedings of the ACM International Conference on Object Oriented Programming Systems Languages and Applications (OOPSLA)*, pages 307–324, 2011.

[45] Felippe Vieira Zacarias, Paul Carpenter, and Vinicius Petrucci. Memory demands in disaggregated HPC: How accurate do we need to be? In *International Workshop on Performance Modeling, Benchmarking and Simulation of High Performance Computer Systems (PMBS)*, 2021.